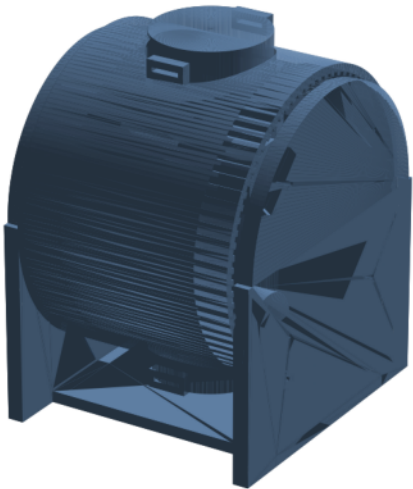


Cross-axis tank tumbler — internal drive

eight wheels inside the drum, bearing outward on grooves cut into its bore

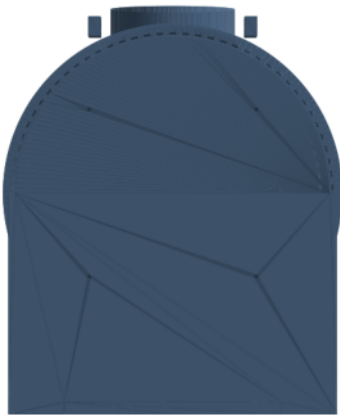
ISO



FRONT



SIDE



TOP



THE PRINCIPLE

A tube lies on its side. A bore runs through it at right angles and the tank lives in that bore, so one turn of the drum is one full inversion of the tank – the motion the Paterson was designed around.

The first 34 mm at each open end carries square grooves on its INNER wall. FOUR wheels per end sit inside the bore, 90° apart, pressing OUTWARD against those grooves. Eight in all.

DRIVE

supply	motor	inversions/min
3.0 V	120	30
4.5 V	185	47
6.0 V	250	63

ratio 31.5 / 125 = 0.252
78 grooves at 10.07 mm pitch
8 motors 4 mm each at 803.5 mm apart

PRINT MANIFEST

part	qty	bbox mm	cm³	filament	fits
assembly	1	232×270×326	7630	3312 g	—
base	1	232×260×300	2282	991 g	OK
drum	1	198×270×290	1916	832 g	OK
motor_mount	8	55×52×27	29	100 g	OK

total printed material ≈ 1922 g

FOUR PER END FULLY CONSTRAINS THE DRUM

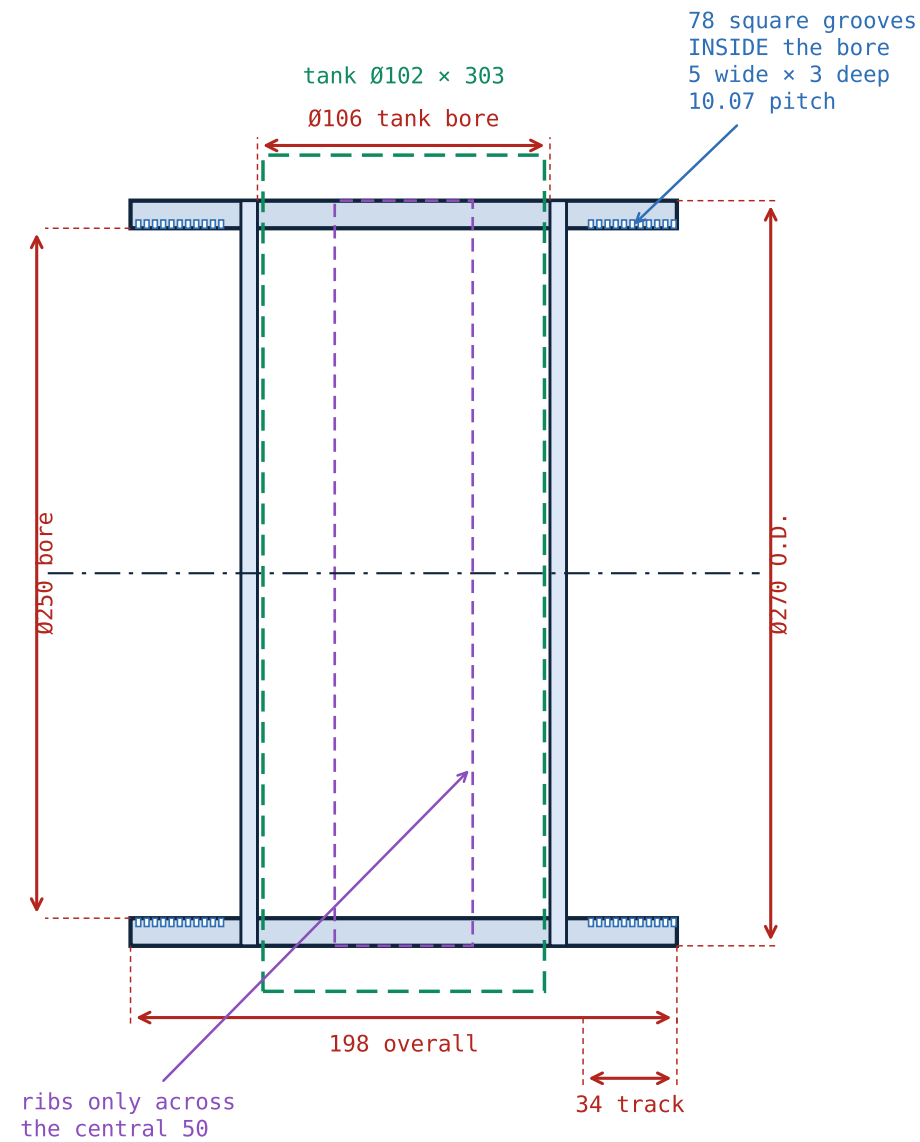
A wheel inside a bore pushes the drum OUTWARD, so the upper pair carry the weight and the lower pair stop it lifting. Four at 90° trap the drum radially in every direction – a proper four-jaw constraint rather than the hanging sheave that two would give, and it doubles the driving contacts.

AXIAL CONSTRAINT

The grooves run axially, so they do not locate the drum along X. The wheel sits in a 34 mm band with 5 mm of end float against the 29 mm tyre – that float is the constraint. Keep it.

Drum

×1 · a tube with grooves inside both ends



SECTION ON THE BORE AXIS



CONSTRUCTION

A plain tube, Ø270 outside and Ø250 inside – 10 mm wall. Grooves cut 3 mm outward from the bore leave 7 mm of wall at the root.

A Ø118 sleeve carries the tank bore across the tube. Four ribs tie it to the shell, but only across the central 50 mm – everything outboard of that is left clear so the motor bodies have somewhere to live.

PRINT IT IN TWO HALVES

On end it is 198 tall and Ø270, which fits – but the Ø106 cross-bore then prints horizontally and needs support down its whole length.

Split on the plane containing both axes: each half prints flat-face-down support-free, and you lay the tank in rather than threading it through.

THE GROOVES ARE THE ONE GUESS

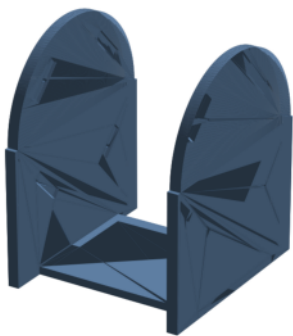
10.07 mm pitch assumes the tread block spacing of the 3766 wheel, which Adafruit does not publish. Measure a real tyre before you commit to 78 of them.

If they will not index, a plain high-friction bore works nearly as well – the preload is doing most of the work either way.

Base and motor mount

two uprights, a flat connector, eight mounts

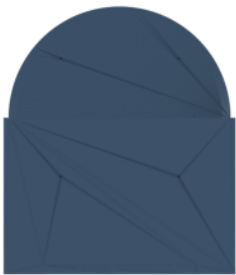
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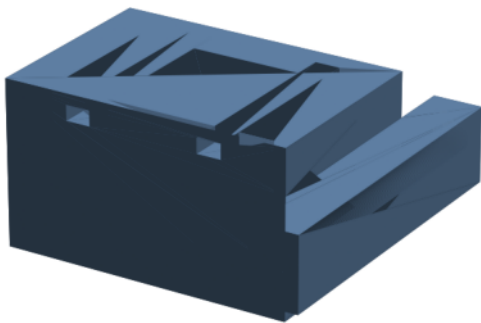
FRONT



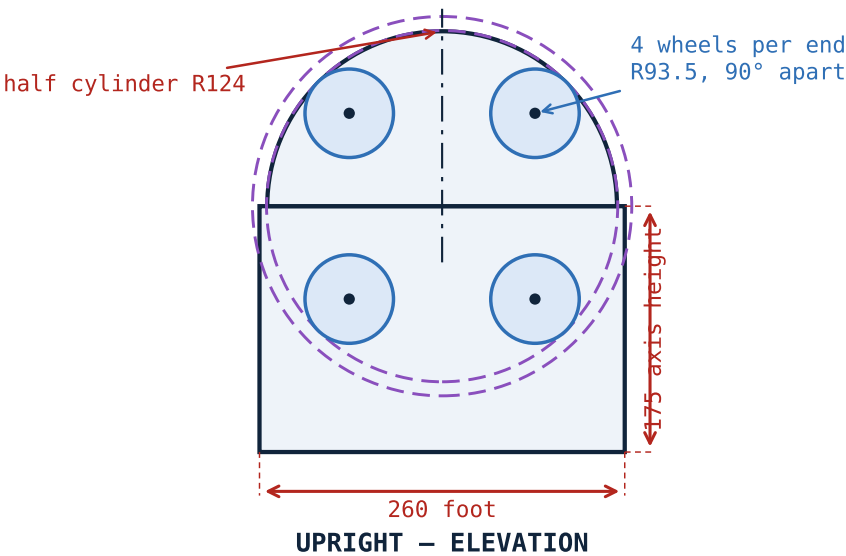
SIDE



ISO



FRONT



BASE

One piece: a flat 10 mm connector between two uprights, each a rectangle with a half cylinder on top. Plates are 14 mm thick and stand 3 mm clear of the drum ends.

Each upright carries FOUR motor mounts on its INBOARD face. Nothing passes through the plate – motor, mount and wheel are all inside the drum bore.

ASSEMBLY: bolt all eight mounts to the uprights first, then lower the drum straight down over them. The bore is Ø250 and the motor cluster spans Ø206, so it drops in clear.

THE ONE CONSTRAINT THE BRIEF COULD NOT HAVE

The motor body is 70 mm long and the groove band is only 34 mm deep, so the motor cannot sit wholly within the band. It is mounted gearbox-outward at the drum mouth with the wheel on its inboard shaft, and the body extends a further 70 mm INTO the drum along its axis.

That is why the ribs stop at the central 50 mm. Measured off the mesh, the bore is clear to R125 everywhere the motors live, and the motor envelope reaches R103 – 22 mm of radial clearance to the rotating wall.

Bill of materials and full view set

and what is still wrong with it

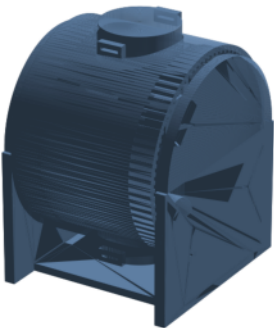
BOUGHT PARTS

part	description	qty	note
Adafruit 3777	TT gearbox motor	8	four per end
Adafruit 3766	TT wheel Ø63 × 29	8	square tread blocks
M4 × 30	preload adjuster	8	+ nuts
M4 × 16	mount to upright	16	+ nuts
Zip tie 2.5 mm	motor retention	16	two per motor
Webbing strap 25 mm	tank retention	2	through the four tabs
PSU + PWM	6 V 10 A	1	8 motors in parallel

REMAINING WEAKNESSES

- WIRING ROTATES PAST THE MOTORS. Four motors sit inside a rotating tube. Their leads must exit through the open end without fouling the drum – route them along the upright, not across the mouth.
- THE DRUM RUNS ON EIGHT TYRES. No bearings anywhere. Silicone on printed plastic sheds dust and creeps; wheels are a consumable and preload needs re-checking.
- EIGHT MOTORS WILL FIGHT EACH OTHER. TT speed spread is about ±10 %, and eight friction-coupled to one drum means the fast ones drive and the slow ones drag. It works, but it costs current and tread life. Stall current is now about 12 A.
- OPEN LOOP. No encoder, so the tank parks wherever it stops. Add a microswitch if it must stop upright for filling.

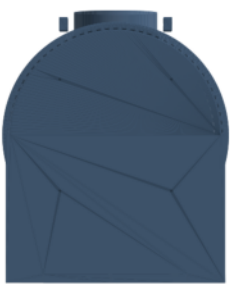
ISO



FRONT



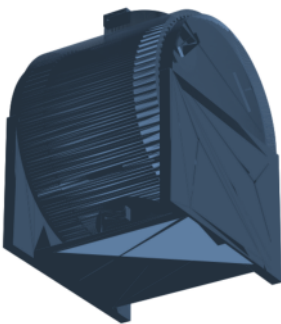
SIDE



TOP



UNDER



ISO-REAR

